

POL 51 Midterm Study Guide

The midterm will be taken virtually (through Canvas on Lockdown browser) and consist of 35 multiple-choice questions (1 point each).

Example question: _____ refers to the extent to which a measurement technique yields the same results when repeated, while _____ refers to the extent to which the measurement accurately measures or captures the underlying concept.

(a) Reliability, validity

(b) Validity, reliability

(c) Internal validity, external validity

(d) External validity, internal validity

1.) Theory-Building

Role of prediction

Behavioralism

Models: generalizability and parsimoniousness (Occam's razor)

Steps in the scientific method (especially theory and hypotheses)

2.) Rational Choice/Game Theory

Rational choice theory

Utility and expected utility

Game theory

Transitivity/intransitivity of preferences

Ordinal vs. cardinal utility

Prisoner's dilemma/tragedy of the commons games

Nash equilibrium

Ultimatum games

The minimax principle

Bounded rationality and heuristics

Anchoring

Prospect theory and risk/loss aversion

Spatial voting model

Median voter theorem

3.) Causation and Research Design

Causality and causal statements

The counterfactual ideal

Confounding variables and spurious relationships

Four hurdles of demonstrating causality

Experimental vs. observational research designs (advantages and disadvantages)

4.) Measurement

Operationalization

Reliability and validity

Measurement error

Three levels of measurement (categorical, ordinal, and continuous)

Central tendency and variation statistics

5.) R Basics

How to create and assign objects

Basic arithmetic commands (addition, subtraction, multiplication, division)

Installing and loading packages

Basic scatterplots

Loading data

The random normal distribution